

# Christoph Junghans

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Group Leader

Applied Computer Science Group (CCS-7)  
Computer, Computational, & Statistical Sciences Division  
Los Alamos National Laboratory, NM, USA

Phone: ++1-505-695-2142

Email: [junghans@lanl.gov](mailto:junghans@lanl.gov)

Web: [www.lanl.gov/junghans](http://www.lanl.gov/junghans)

## Research Interests

- **Currently:** Applied computer science
- Software design and management – core developer of the coarse-graining package [VOTCA](#)
- Porting and optimization of scientific codes ([VOTCA](#), [Cabana](#), [FleCSI](#))
- Novel software development workflows for Exascale computing
- Systematic coarse-graining of liquids and implementation of new methodologies
- Development of multiscale and multiphysics simulation approaches (incl. AI)

## Education

- Oct 2010 **Dr. rer. nat. (Ph.D.) in Physics**, *magna cum laude*,  
Johannes Gutenberg University, Mainz, Germany  
[Between the Scales: Water from different Perspectives](#)  
An adaptive resolution molecular dynamics study using various coarse-graining methods
- Nov 2006 **Diploma (M.Sc.) in Physics**, *summa cum laude*, University of Leipzig, Germany  
[Aggregation of Mesoscopic Protein-like Heteropolymers](#)  
A generalized ensemble Monte Carlo study using multicanonical techniques

## Work Experience

- Feb 2021 - Present **Group Leader**, Applied Computer Science Group, Computer, Computational, and Statistical Sciences Division, Los Alamos National Lab
- Feb 2016 - Present **CNLS Affiliate**, Center for Nonlinear Studies, Theoretical Division, Los Alamos National Lab
- July 2018 - Feb 2021 **Deputy Group Leader**, Applied Computer Science Group, Computer, Computational, and Statistical Sciences Division, Los Alamos National Lab
- Jan 2017 - July 2018 **Deputy Team Leader**, Co-Design Team, Computer, Computational, and Statistical Sciences Division, Los Alamos National Lab
- May 2014 - July 2018 **Staff Scientist**, Computer, Computational, and Statistical Sciences Division, Los Alamos National Lab
- Mar 2013 – Apr 2014 **Director’s Postdoc. Fellow**, Theoretical Division, Los Alamos National Lab
- Nov 2011 – Feb 2013 **Postdoc. Research Assistant**, Theoretical Division, Los Alamos National Lab
- May 2012 – June 2012 **Member**, The Kavli Institute for Theoretical Physics, University of California, Santa Barbara, CA
- Nov 2010 – Oct 2011 **Postdoc. Research Assistant**, Polymer Theory Group, Max Planck Institute for Polymer Research, Mainz, Germany
- Jan 2007 – Oct 2010 **Ph.D. Student**, Polymer Theory Group, Max Planck Institute for Polymer Research, Mainz, Germany
- Jan 2009 – July 2009 **Specialist for Application Performance & Deep Computing**, Internship, IBM Systems & Technology Group Europe
- Oct 2003 – Sept 2006 **Student Assistant**, Institute for Theoretical Physics, University of Leipzig
- Aug 2005 – Oct 2005 **Research Student** in the “Scientific Computing” program, Jülich Supercomputing Centre, Germany

Sep 2004                      **Student Assistant**, Institute for Meteorology, University of Leipzig  
 Feb 2003 – April 2003    **Research Student**, Chair for Fluid Mechanics, Martin Luther University of Halle-Wittenberg, Germany

## Open Source Projects

|                           |                                                             |                      |
|---------------------------|-------------------------------------------------------------|----------------------|
| <a href="#">VOTCA</a>     | Versatile Toolkit for coarse-graining applications          | Core Developer       |
| <a href="#">Cabana</a>    | Performance-portable library for particle-based simulations | Core Developer       |
| <a href="#">FleCSI</a>    | Flexible Computer Science Infrastructure                    | Developer            |
| <a href="#">Fedora</a>    | Linux distribution                                          | Contributor          |
| <a href="#">OpenSuse</a>  | Linux distribution                                          | Contributor          |
| <a href="#">ESPReSo++</a> | Successor of the ESPResSo simulation package                | Developer            |
| <a href="#">GroMaCS</a>   | Versatile package to perform molecular dynamics             | Developer (inactive) |
| <a href="#">ESPReSo</a>   | Extensible Simulation Package for Research on Soft matter   | Developer (inactive) |

More information can be found on [my GitHub profile](#) and [my OpenHUB profile](#).

## Awards

|           |                                                                                          |
|-----------|------------------------------------------------------------------------------------------|
| Nov 2019  | Top 10% most cited PLOS ONE papers published in 2015                                     |
| Sep 2017  | LAAP Award in recognition of outstanding contributions to the ISTI Summer School Program |
| Feb 2017  | Postdoc. Distinguished Mentor Award, LANL                                                |
| July 2016 | LAAP Achievement Award in appreciation for active service on the WSST team               |
| Jan 2015  | Classified as outstanding researcher, USCIS                                              |
| Jan 2014  | Most cited article published in EPJE in the past 5 years                                 |
| Mar 2013  | Los Alamos National Laboratory Director's Post-Doctoral Fellowship                       |
| May 2012  | Member of the Kavli Institute for Theoretical Physics                                    |
| June 2010 | Participant of the 60th Lindau Nobel Laureate Meeting                                    |
| Nov 2006  | Diploma with honor (highest possible grade)                                              |
| Nov 2005  | Wolfgang Natonek award, University of Leipzig                                            |
| Oct 2003  | Teubner award, Department for Physics and Earth Science, University of Leipzig           |
| July 2000 | Book award of the German Physical Society (DPG) for high school graduates                |

## Professional Skills

### *Managerial*

- Line management, since July 2018, Group of 65+ staff scientists and Postdocs, \$50M+ budget
- Team management, 2 years, technical leadership of 10+ team member
- Project management, multiple lead roles in Exascale Computing Project (ECP), Laboratory Directed Research & Development (LDRD) and Advanced Simulation & Computing (ASC) projects

### *Referee for Scientific Journals*

- ACS: J. Chem. Inf. Model. & J. Phys. Chem.
- AIP: J. Chem. Phys.
- Elsevier: Comp. Phys. Comm.
- RSC: Phys. Chem. Chem. Phys.
- and others

*Code Review*

- [Gromacs](#)
- [OpenSuse Build Service](#)

*Conferences*

- Member of the Organization Committee for the [Salishan Conference on High Speed Computing](#)
- Chairman for APS annual meetings

*Representative*

|                       |                                                                                                                                                                                                                                                             |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Nov 2018 - Sep 2022   | Member of the Information Science and Technology Institute (ISTI) council, LANL                                                                                                                                                                             |
| July 2015 - June 2016 | Member of the Worker Safety & Security Team (WSST) for CCS-7, LANL                                                                                                                                                                                          |
| July 2007 - Dec 2008  | PhD representative of the theory group, Max Planck Institute for Polymer Research, Mainz                                                                                                                                                                    |
| Nov 2007 - Nov 2008   | PhD representative of the MPI for Polymer Research                                                                                                                                                                                                          |
| Oct 2002 - Sep 2005   | Elected member of student government (Fachschaftratsrat) of the Department for Physics and Earth Science, University of Leipzig - Extensive committee work in this period incl. spokesperson, treasurer and member of several faculty/university committees |

**Computer Skills**

|                         |                                            |
|-------------------------|--------------------------------------------|
| <b>General</b>          | Linux, AIX, Mac OS, DOS, Windows           |
| <b>Programming</b>      | C, Fortran, MPI, OpenMP, C++               |
| <b>Scripting</b>        | Shell, awk, Perl, expect, PHP, tcl, Python |
| <b>Markup Languages</b> | HTML, latex, mediawiki, txt2tags           |

**Teaching Experience**

|                      |                                                                                                                                                                                  |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Aug 2021             | “VOTCA and LAMMPS for electronic spectroscopy and transport” (Tutorial), <a href="#">LAMMPS Workshop and Symposium</a> , Virtual                                                 |
| Aug 2019             | “VOTCA and LAMMPS for electronic spectroscopy and transport” (Tutorial), <a href="#">LAMMPS Workshop and Symposium</a> , Albuquerque, NM                                         |
| May 2018 - Aug 2018  | <a href="#">Co-Design Summer School</a> (6 Students), Los Alamos National Laboratory                                                                                             |
| May 2017 - Jul 2017  | <a href="#">Co-Design Summer School</a> (6 Students), Los Alamos National Laboratory                                                                                             |
| Oct 2016             | “Coarse-Graining with VOTCA” (Tutorial), <a href="#">CECAM Workshop “Multiscale Simulations of Soft Matter with Hands-On Tutorials on ESPResSo++ and VOTCA”</a> , Mainz, Germany |
| Jun 2016 - Aug 2016  | <a href="#">Co-Design Summer School</a> (7 Students), Los Alamos National Laboratory                                                                                             |
| May 2016             | Gitlab Hands-On course, ASC/NGC Program, LANL                                                                                                                                    |
| Oct 2015             | “Coarse-Graining with VOTCA” (Tutorial), <a href="#">CECAM Summer School “Simulating Soft and Active Matter with ESPResSo, ESPResSo++ and VOTCA”</a> , ICP Stuttgart             |
| Sep 2015             | Git Hands-On course, ASC/NGC Program, LANL                                                                                                                                       |
| Jun 2015 - Aug 2015  | <a href="#">Co-Design Summer School</a> (6 Students), Los Alamos National Laboratory                                                                                             |
| Nov 2014             | Git IC Tutorial, Institutional Computing, LANL                                                                                                                                   |
| Oct 2014             | Hands-on: VOTCA, <a href="#">CECAM “School on Multiscale Modeling and Use of Espresso++ and VOTCA”</a> , Mainz, Germany                                                          |
| Jun 2014 - Aug. 2014 | <a href="#">Co-Design Summer School</a> (6 Students), Los Alamos National Laboratory                                                                                             |
| Aug 2014             | Gromacs Tutorial, <a href="#">The Eighth q-bio Summer School</a> , Albuquerque, NM (LA-UR 14-26188)                                                                              |

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|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Jun 2014            | Gromacs IC Tutorial, Institutional Computing, LANL (LA-UR 14-24814)                                                                                                  |
| Oct 2013            | “Systematic Coarse-Graining with VOTCA” (Tutorial), <a href="#">CECAM Summer School “Simulating Soft Matter with ESPResSo, ESPResSo++ and VOTCA”</a> , ICP Stuttgart |
| Jul 2013 - Aug 2013 | <a href="#">Co-Design Summer School</a> (6 Students), Los Alamos National Laboratory                                                                                 |
| Jul 2013            | Gromacs Tutorial, <a href="#">The Seventh q-bio Summer School</a> , Santa Fe, NM                                                                                     |
| Oct 2012            | “Systematic Coarse-Graining with VOTCA” (Tutorial), <a href="#">CECAM Workshop “Simulating Soft Matter with ESPResSo, ESPResSo++ and VOTCA”</a> , ICP Stuttgart      |
| Jul 2012            | Gromacs Tutorial, <a href="#">The Sixth q-bio Summer School</a> , Santa Fe, NM                                                                                       |
| Oct 2011            | Hands-on: VOTCA, <a href="#">CECAM Workshop “Coarse-grained Simulation of Biological Soft Matter Systems using ESPResSo”</a> , ICP Stuttgart                         |
| May 2010            | Votca Workshop, <a href="#">CSI Darmstadt</a>                                                                                                                        |
| Apr 2008 - Sep 2008 | Theoretical physics III, Johannes Gutenberg University of Mainz                                                                                                      |
| Oct 2007 - Mar 2008 | Theoretical physics II, Johannes Gutenberg University of Mainz                                                                                                       |
| Apr 2007 - Sep 2007 | Theoretical physics I, Johannes Gutenberg University of Mainz                                                                                                        |
| Apr 2006 - Sep 2006 | Introduction to computer simulations I, University of Leipzig                                                                                                        |
| Sep 2004 - Oct 2004 | Mathematical preparation course for first-year students, University of Leipzig                                                                                       |

## Mentoring

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|---------------------|-----------------------------------------------------------------------------------|
| Feb 2022 - Oct 2023 | Postdoc. Research Assistant (now Staff Scientist), Los Alamos National Laboratory |
| Jun 2019 - Mar 2021 | Postdoc. Research Assistant, Los Alamos National Laboratory                       |
| Feb 2018 - Apr 2020 | Postdoc. Research Assistant (now Staff Scientist), Los Alamos National Laboratory |
| May 2018 - Feb 2019 | Postdoc. Research Assistant (now Staff Scientist), Los Alamos National Laboratory |
| Feb 2017 - Jan 2019 | Postdoc. Research Assistant (now Staff Scientist), Los Alamos National Laboratory |
| Sep 2018            | Member of PhD committee, University of Reims Champagne-Ardenne                    |
| Jun 2018 - Aug 2018 | ECP Summer Student, Los Alamos National Laboratory                                |
| May 2018 - Aug 2018 | Co-Design Summer School (6 Students), Los Alamos National Laboratory              |
| May 2018 - Aug 2018 | ASC Summer Student, Los Alamos National Laboratory                                |
| Feb 2016 - May 2018 | Postdoc. Research Assistant (now Staff Scientist), Los Alamos National Laboratory |
| Dec 2016 - Apr 2018 | Postdoc. Research Assistant (now Staff Scientist), Los Alamos National Laboratory |
| Jun 2017 - Aug 2017 | High School Internship Student, Los Alamos National Laboratory                    |
| Apr 2017 - Jun 2017 | ASC Summer Student, Los Alamos National Laboratory                                |
| May 2017 - Jul 2017 | Data-Science at Scale Summer Student, Los Alamos National Laboratory              |
| May 2017 - Jul 2017 | Co-Design Summer School (6 Students), Los Alamos National Laboratory              |
| May 2015 - May 2017 | Metropolis Postdoc. Fellow (now Staff Scientist), Los Alamos National Laboratory  |
| Aug 2016 - Apr 2017 | Postdoc. Research Assistant (now Staff Scientist), Los Alamos National Laboratory |
| Jul 2016 - Sep 2016 | CNLS Summer Student, Los Alamos National Laboratory                               |
| Jun 2016 - Aug 2016 | Co-Design Summer School (7 Students), Los Alamos National Laboratory              |
| Jul 2015 - Sep 2015 | IC Summer Student, Los Alamos National Laboratory                                 |
| Jun 2015 - Aug 2015 | Co-Design Summer School (6 Students), Los Alamos National Laboratory              |
| Jul 2014 - Sep 2014 | CNLS Summer Student, Los Alamos National Laboratory                               |
| Jun 2014 - Aug 2014 | Co-Design Summer School (6 Students), Los Alamos National Laboratory              |
| Jul 2013 - Aug 2013 | Co-Design Summer School (6 Students), Los Alamos National Laboratory              |
| Jun 2010 - Aug 2010 | Summer Student, University of Mainz                                               |
| Oct 2005 - Mar 2006 | Simulation methods in generalized ensembles (2 Students), University of Leipzig   |

## Talks

Over 50 invited and contributed talks

### *Selected Talks*

- Sep. 2018 Lecture: “Co-Design at Los Alamos National Laboratory”, [University of Reims Champagne-Ardenne](#)
- Oct. 2016 Talk: “Recent developments in VOTCA”, [CECAM Workshop “Multiscale Simulations of Soft Matter with Hands-On Tutorials on ESPResSo++ and VOTCA”](#), Mainz, Germany
- May 2012 Seminar: “Particle-based multi-scale simulations using the Adaptive Resolution Scheme”, [IBM Research, Almaden](#)
- July 2011 Colloquium: “Locality Analysis via Adaptive Resolution Simulations”, Theoriekolloquium, [Faculty of Natural Sciences II, Martin Luther University Halle-Wittenberg](#)

## Publications

h-index: [25 \(Google Scholar\)](#), [23 \(ResearcherID\)](#)

### *Reviewed Papers*

47. J. M. Miller, D. A. Holladay, J. H. Peterson, C. M. Mauney, R. Berger, A. Pietarila Graham, K. C. Tsai, B. Barker, A. Holas, A. E. Mattsson, M. Gogilashvili, J. C. Dolence, C. D. Meyer, S. Swaminarayan, and **C. Junghans**, *Singularity-EOS: Performance Portable Equations of State and Mixed Cell Closures*, [J. Open Source Softw.](#) **9**, 6805 (2024).
46. B. Baumeier, J. Wehner, N. Renaud, F. Zapata Ruiz, R. Halver, P. Madhikar, R. Gerritsen, G. Tirimbo, J. Sijen, D. Rosenberger, J. S. Brown, V. Sundaram, J. Krajniak, M. Bernhardt, and **C. Junghans**, *VOTCA: multiscale frameworks for quantum and classical simulations in soft matter*, [J. Open Source Softw.](#) **9**, 6864 (2024).
45. S. T. Reeve, J.-L. Fattebert, S. DeWitt, D. Joy, P. Seleson, S. Slattery, A. Scheinberg, R. Halver, **C. Junghans**, C. F. A. Negre, M. E. Wall, Y. Zhang, A. M. Niklasson, D. Perez, S. M. Mniszewski, and J. Belak, *Co-design for Particle Applications at Exascale*, [Comp. Sci. Eng. PrePrints](#)(2024).
44. S. Karra, M. Mehana, N. Lubbers, Y. Chen, A. Diaw, J. E. Santos, A. Pachaliev, R. S. Pavel, J. R. Haack, M. McKerns, **C. Junghans**, Q. Kang, D. Livescu, T. C. Germann, and H. S. Viswanathan, *Predictive Scale-Bridging Simulations through Active Learning*, [Sci. Rep.](#) **13**, 16262 (2023).
43. S. Wijesinghe, **C. Junghans**, D. Perahia, and G. S. Grest, *Polydots, soft nanoparticles, at membrane interfaces*, [RSC Adv.](#) **13**, 19227 (2023).
42. R. Halver, **C. Junghans**, and G. Sutmann, *Using heterogeneous GPU nodes with a Cabana-based implementation of MPCD*, [Paral. Comp.](#) **117**, 103033 (2023).
41. J. Vance, Z.-H. Xu, N. Tretyakov, T. Stuehn, M. Rampp, S. Eibl, **C. Junghans**, and A. Brinkmann, *Code modernization strategies for short-range non-bonded molecular dynamics simulations*, [Comp. Phys. Comm.](#) **290**, 108760 (2023).
40. R. Halver, **C. Junghans**, and G. Sutmann, *Kokkos-Based Implementation of MPCD on Heterogeneous Nodes*, in: R. Wyrzykowski, J. Dongarra, E. Deelman, and K. Karczewski (Eds.), *Parallel Processing and Applied Mathematics. PPAM 2022. Lecture Notes in Computer Science*, [Vol. 13827](#) (2023).

39. A. Pachalieva, R. S. Pavel, J. E. Santos, A. Diaw, N. Lubbers, M. Mehana, J. R. Haack, H. S. Viswanathan, D. Livescu, T. C. Germann, and **C. Junghans**, *GLUE Code: A framework handling communication and interfaces between scales*, [\*J. Open Source Softw.\* \*\*7\*\*, 4822 \(2022\)](#).
38. P. Grete, J. C. Dolence, J. M. Miller, J. Brown, B. Ryan, A. Gaspar, F. Glines, S. Swaminarayan, J. Lippuner, C. J. Solomon, G. Shipman, **C. Junghans**, D. Holladay, and J. M. Stone, *Parthenon - a performance portable block-structured adaptive mesh refinement framework*, [\*Int. J. High Perf. Comp. App.\* \*\*37\*\*, 465 \(2023\)](#).
37. S. Slattery, S. T. Reeve, **C. Junghans**, D. Lebrun-Grandie, R. Bird, G. Chen, S. Fogerty, Y. Qiu, S. Schulz, A. Scheinberg, A. Isner, K. Chong, S. Moore, T. Germann, J. Belak, and S. Mniszewski, *Cabana: A Performance Portable Library for Particle-Based Simulations*, [\*J. Open Source Softw.\* \*\*7\*\*, 4115 \(2022\)](#).
36. J. Bakosi, R. Bird, F. Gonzalez, **C. Junghans**, W. Li, H. Luo, A. Pandare, and J. Waltz, *Asynchronous distributed-memory task-parallel algorithm for compressible flows on unstructured 3D Eulerian grids*, [\*Adv. Eng. Softw.\* \*\*160\*\*, 102962 \(2021\)](#).
35. S. M. Mniszewski, J. Belak, J.-L. Fattbert, C. F. A. Negre, S. R. Slattery, A. A. Adedoyin, R. F. Bird, C.S. Chang, G. Chen, S. Ethier, S. Fogerty, S. Habib, **C. Junghans**, D. Lebrun-Grandie, J. Mohd-Yusof, S. G. Moore, D. Osei-Kuffuor, S. J. Plimpton, A. Pope, S. T. Reeve, L. Ricketson, A. Scheinberg, A. Y. Sharma, and M. E. Wall, *Enabling Particle Applications for Exascale Computing Platforms*, [\*Int. J. High Perf. Comp. App.\* \*\*35\*\*, 572 \(2021\)](#).
34. A. Diaw, K. Barros, J. Haack, **C. Junghans**, B. Keenan, Y. W. Li, D. Livescu, N. Lubbers, M. McKerns, R. S. Pavel, D. Rosenberger, I. Sagert, and T. C. Germann, *Multiscale simulation of plasma flows using active learning*, [\*Phys. Rev. E\* \*\*102\*\*, 023310 \(2020\)](#).
33. N. Lubbers, A. Agarwal, Y. Chen, S. Son, M. Mehana, Q. Kang, S. Karra, **C. Junghans**, T. C. Germann, and H. S. Viswanathan, *Modeling and scale-bridging using machine learning: nanoconfinement effects in porous media*, [\*Sci. Rep.\* \*\*10\*\*, 13312 \(2020\)](#).
32. G. Tirimbo, V. Sundaram, O. Caylak, W. Scharpach, J. Sijen, **C. Junghans**, J. Brown, F. Zapata Ruiz, N. Renaud, J. Wehner, and B. Baumeier, *Excited-State Electronic Structure of Molecules Using Many-Body Green's Functions: Quasiparticles and Electron-Hole Excitations with VOTCA-XTP*, [\*J. Chem. Phys.\* \*\*152\*\*, 114103 \(2020\)](#).
31. H. V. Guzman, N. Tretyakov, H. Kobayashi, A. C. Fogarty, K. Kreis, J. Krajniak, **C. Junghans**, K. Kremer, and T. Stuehn, *ESPReso++ 2.0: Advanced methods for multiscale molecular simulation*, [\*Comp. Phys. Comm.\* \*\*238\*\*, 66 \(2019\)](#).
30. J. Wehner, L. Brombacher, J. Brown, **C. Junghans**, O. Caylak, Y. Khalak, P. Madhikar, G. Tirimbo, and B. Baumeier, *Electronic Excitations in Complex Molecular Environments: Many-Body Green's Functions Theory in VOTCA-XTP*, [\*J. Chem. Theo. Comp.\* \*\*14\*\*, 6253 \(2018\)](#).
29. C. Krekeler, A. Agarwal, **C. Junghans**, M. Praprotnik, and L. Delle Site, *Adaptive Resolution Molecular Dynamics Technique: Down to the Essential*, [\*J. Chem. Phys.\* \*\*149\*\*, 024104 \(2018\)](#).
28. H. V. Guzman, **C. Junghans**, K. Kremer, and T. Stuehn, *Scalable and fast heterogeneous molecular simulation with predictive parallelization schemes*, [\*Phys. Rev. E\* \*\*96\*\*, 053311 \(2017\)](#).
27. S. K. Gutierrez, K. Davis, D. Arnold, R. S. Baker, R. W. Robey, P. McCormick, D. Holladay, J. A. Dahl, J. Zerr, F. Weik, and **C. Junghans**, *Accommodating Thread-Level Heterogeneity in Coupled Parallel Applications*, in: *Proceedings of 2017 IEEE International Parallel and Distributed Processing Symposium (IPDPS)*, Orlando, FL, USA, 469 (2017).



26. **C. Junghans**, A. Agarwal, and L. Delle Site, *Computational efficiency and Amdahl's law for the adaptive resolution simulation technique*, *Comp. Phys. Comm.* **215**, 20 (2017).
25. T. E. de Oliveira, P. A. Netz, K. Kremer, **C. Junghans**, and D. Mukherji, *C-IBI: Targeting cumulative coordination within an iterative protocol to derive coarse-grained models of (multi-component) complex fluids*, *J. Chem. Phys.* **144**, 174106 (2016).
24. R. S. Pavel, A. L. McPherson, T. C. Germann, and **C. Junghans**, *Database Assisted Distribution to Improve Fault Tolerance for Multiphysics Applications*, in: *Proceedings of the 2nd International Workshop on Hardware-Software Co-Design for High Performance Computing 2015 (Co-HPC '15)*, ACM, New York, NY, USA, [Article 4](#).
23. S. Y. Mashayak, M. N. Jochum, K. Koschke, N. R. Aluru, V. Rühle, and **C. Junghans**, *Relative entropy and optimization-driven coarse-graining methods in VOTCA*, *PLoS ONE* **10**, e131754 (2015). "Among the top 10% most cited PLoS ONE papers published in 2015." (2019)
22. S. M. Mniszewski, **C. Junghans**, A. F. Voter, D. Perez, and S. J. Eidenbenz, *TADSim: Discrete Event-based Performance Prediction for Temperature Accelerated Dynamics*, *Trans. Mod. Comp. Sim.* **25**, 15 (2015).
21. D. G. Roehm, R. S. Pavel, K. Barros, B. Rouet-Leduc, A. L. McPherson, T. C. Germann, and **C. Junghans**, *Distributed Database Kriging for Adaptive Sampling*, *Comp. Phys. Comm.* **192**, 138 (2015).
20. S. Bevc, **C. Junghans**, and M. Praprotnik, *STOCK: Structure Mapper and Online Coarse-Graining Kit for Molecular Simulations*, *J. Comp. Chem.* **36**, 467 (2015).
19. **C. Junghans**, D. Perez and T. Vogel, *Molecular Dynamics in the Multicanonical Ensemble: Equivalence of Wang-Landau Sampling, Statistical Temperature Molecular Dynamics, and Metadynamics*, *J. Chem. Theo. Comp.* **10**, 1843 (2014).
18. B. Rouet-Leduc, K. Barros, E. Cieren, V. Elango, **C. Junghans**, T. Lookman, J. Mohd-Yusof, R. S. Pavel, A. Y. Rivera, D. Roehm, A. L. McPherson, T. C. Germann, *Spatial adaptive sampling in multiscale simulation*, *Comp. Phys. Comm.* **185**, 1857 (2014).
17. A. Nagarajan, **C. Junghans** and S. Matysiak, *Multiscale simulation of liquid water using a four-to-one mapping for coarse-graining*, *J. Chem. Theo. Comp.* **9**, 5168 (2013).
16. S. Bevc, **C. Junghans**, K. Kremer and M. Praprotnik, *Adaptive resolution simulation of salt solutions*, *New J. Phys.*, **15**, 105007 (2013).
15. P. Ganguly, D. Mukherji, **C. Junghans** and N. F. A. van der Vegt, *Kirkwood-Buff coarse-grained force fields for aqueous solutions*, *J. Chem. Theo. Comp.*, **8**, 1802 (2012).
14. S. Fritsch, S. Poblete, **C. Junghans**, G. Ciccotti, L. Delle Site and K. Kremer, *Adaptive resolution molecular dynamics simulation through coupling to an internal particle reservoir*, *Phys. Rev. Lett.* **108**, 170602 (2012).
13. S. Fritsch, **C. Junghans** and K. Kremer, *Structure formation of toluene around C60: Implementation of the Adaptive Resolution Scheme (AdResS) into GROMACS*, *J. Chem. Theo. Comp.* **8**, 398 (2012).
12. V. Rühle and **C. Junghans**, *Hybrid approaches to coarse-graining using the VOTCA package: liquid hexane*, *Macromol. Theory Simul.*, **20**, 472 (2011).
11. **C. Junghans**, W. Janke and M. Bachmann, *Hierarchies in Nucleation Transitions*, *Comp. Phys. Comm.*, **182**, 1937 (2011).

10. B. P. Lambeth, Jr., **C. Junghans**, K. Kremer, C. Clementi, and L. Delle Site, *Communication: On the Locality of Hydrogen Bond Networks at Hydrophobic Interface*, *J. Chem. Phys.* **133**, 221101 (2010).
9. **C. Junghans** and S. Poblete, *A reference implementation of the adaptive resolution scheme in ESPResSo*, *Comp. Phys. Comm.* **181**, 1449 (2010).
8. V. Rühle, **C. Junghans**, A. Lukyanov, K. Kremer and D. Andrienko, *Versatile Object-oriented Toolkit for Coarse-graining Applications*, *J. Chem. Theo. Comp.* **5**, 3211 (2009).
7. **C. Junghans**, M. Bachmann and W. Janke, *Statistical Mechanics of Aggregation and Crystallization for Semiflexible Polymers*, *Europhys. Lett.* **87**, 40002 (2009).
6. H. Wang, **C. Junghans** and K. Kremer, *Comparative atomistic and coarse-grained study of water: What do we lose by coarse-graining?*, *Euro. Phys. J. E* **28**, 221 (2009). “Most cited article published in the journal in the past 5 years” (2014)
5. M. Praprotnik, **C. Junghans**, L. Delle Site and K. Kremer, *Simulation approaches to soft matter: Generic statistical properties vs. chemical details*, *Comp. Phys. Comm.* **179**, 51 (2008).
4. **C. Junghans**, M. Bachmann and W. Janke, *Thermodynamics of Peptide Aggregation Processes: An Analysis from Perspectives of Three Statistical Ensembles*, *J. Chem. Phys.* **128**, 085103 (2008) .
3. **C. Junghans**, M. Praprotnik and K. Kremer, *Transport properties controlled by a thermostat: An extended dissipative particle dynamics thermostat*, *Soft Matter* **4**, 156 (2008).
2. **C. Junghans**, M. Bachmann and W. Janke, *Microcanonical Analyses of Peptide Aggregation Processes*, *Phys. Rev. Lett.* **97**, 218103 (2006).
1. **C. Junghans** and U. H. E. Hansmann, *Numerical Comparison of Wang Landau Sampling and Parallel Tempering for Met-enkephalin*, *Int. J. Mod. Phys. C* **17**, 817 (2006).

More information can be found on [my ResearcherID profile](#), [my Google Scholar profile](#), [my ORCID profile](#), [my ResearchGate profile](#) or [my Impactstory profile](#).

#### Book Chapters

3. L. Delle Site, A. Agarwal, **C. Junghans**, and H. Wang, *Adaptive Resolution Simulation as a Grand Canonical Molecular Dynamics Scheme: Principles, Applications, Perspectives*, in: C. Clementi (Ed.), *Multiscale Methods in Molecular Biophysics*, Series in Computational Biophysics, CRC Press (2016), ISBN 1482225700.
2. E. Apol, R. Apostolov, H.J.C. Berendsen, A. van Buuren, P. Bjelkmar, R. van Drunen, A. Feenstra, S. Fritsch, G. Groenhof, **C. Junghans**, P. Kasson, P. Larsson, P. Meulenhoff, T. Murtola, S. Pall, S. Pronk, R. Schulz, M. Shirts, A. Sijbers, P. Tieleman, B. Hess, D. van der Spoel, and E. Lindahl, *Gromacs User Manual Version 4.6*, [gromacs.org](http://gromacs.org) (2013).
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#### Other Publications

6. *Balancing the load: Los Alamos researchers develop code to distribute computation more efficiently and across increasing numbers of supercomputer processors*. [ASCR Discovery Story](#), [HPCwire](#), [insideHPC](#) and LANL Science Brief (2018).



5. **C. Junghans**, A. K. Hüttel and U. Müller, *Gentoo Linux: Quelltexte und Rolling Releases*, [c't Magazin 16](#), 162 (2012).
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2. **C. Junghans**, M. Bachmann and W. Janke, *Phase Separation in Peptide Aggregation Processes - Multicanonical Study of a Mesoscopic Model*, in: U. H. E. Hansmann, J. Meinke, S. Mohanty and O. Zimmermann (Eds.), *From Computational Biophysics to Systems Biology 2007*, NIC Series Vol. 36, Jülich (2007), 169.
1. **C. Junghans** and U. H. E. Hansmann, *Cross-Check Methods in Protein Simulations*, in: U. H. E. Hansmann, J. Meinke, S. Mohanty and O. Zimmermann (Eds.), *From Computational Biophysics to Systems Biology 2006*, NIC Series Vol. 34, Jülich (2006), 157.